# INTRODUCTION

The Gym Management Application is a groundbreaking solution designed to revolutionize fitness experiences through the integration of Artificial Intelligence (AI). Developed for Android using React Native and powered by Firebase for the backend, this application offers users a comprehensive and interactive platform to achieve their health and wellness goals. It provides personalized AI-generated feedback on workouts, enabling users to track and improve their performance effectively. With real-time interaction with fitness experts and intelligent notifications tailored to individual routines, the application optimizes fitness journeys while promoting healthy habits. This innovative approach bridges technology and wellness, creating a seamless and engaging fitness experience for users.

## 2.1. SYSTEM STUDY

System study refers to the process of examining a situation with the intent of improving it through better process and methods. System study is, therefore the process of gathering and interpreting facts, diagnosing problem and using the information to recommend changes in the system, in other words it means a detailed explanation of description. Before computerizing a system under consideration, it must be analysed. We need to study how it functions currently, what are problems and what are the requirements that the proposed software should meet.

The main components of making software are:

- System and software requirements analysis
- Design and implementation of software
- Ensuring, verifying and maintaining software integrity

The Online Art Gallery Management System provides user friendly interface which can be operated by anyone with little knowledge about the computer system. It stores the information needed by the shop in a database which can be accessed by the administrator, staff and customer. It should maintain a well organized database for storing the information regarding the shop. This helps to eliminate the storage of invalid data.

### 2.1.1 Existing System

The current gym management systems primarily focus on administrative functionalities such as membership management, attendance tracking, and payment processing. While they streamline operations for gym owners and staff, they often lack advanced features to enhance the user experience. Fitness tracking applications available in the market typically require manual input of workout details, making them less efficient and less personalized. Real-time interaction with fitness experts is rarely integrated, and notifications are generic rather than tailored to individual needs. Moreover, existing systems do not effectively leverage AI to provide personalized feedback or adaptive health recommendations, leaving a gap in providing a truly engaging and interactive fitness journey for users.

### 2.1.2 Proposed System

The proposed Gym Management Application is a cutting-edge solution designed to elevate the fitness experience by integrating advanced Artificial Intelligence (AI) technology. Built for Android using React Native and powered by Firebase, the application offers a suite of innovative features that go beyond traditional gym management systems. It provides personalized AI-generated feedback on workouts, enabling users to improve their form, efficiency, and overall performance. The application also facilitates real-time interaction with fitness experts, allowing users to seek guidance, participate in live sessions, and review their progress with certified trainers. Intelligent notifications are tailored to individual needs, offering reminders, recovery tips, and motivational insights to encourage consistency and healthy habits. Additionally, the app ensures seamless fitness tracking by automatically recording workout details, integrating with wearable devices, and delivering comprehensive progress reports. With an intuitive interface and access to customized workout plans and expert advice, this system redefines fitness management by creating an engaging, interactive, and goal-oriented experience for users and gym operators alike.

### 2.1.3. Feasibility Study

Feasibility study is made to see if the project on completion will serve the purpose of the organization for the amount of work, time and effort spent on it. Feasibility study lets the developer foresee the future of the project and its usefulness. Finding out whether a new system is required or not. The study is carried out to the best system that meet performance requirement. This entails identification, description and evaluation of candidate system and selection of the best system for the best system for the job. It simply identifies whether the proposed system is feasible to the organization or not.

There are three aspects in the feasibility study portion of the preliminary investigation

- Technical feasibility
- Economic feasibility
- Operational feasibility

#### 2.1.3.1. Technical Feasibility

The system must be evaluated from technical viewpoint first. The assessment of this feasibility must be based on outline design of the system requirement in the terms of input, output, programs and procedure having identified an outline system, the investigation must go on to suggest the type of equipment, required method of developing the system, method of running the system once it has been designed. The project should be developed such that the necessary functions and performance are achieved within the constraints. The project is developed with latest technology. There are only minimal constraints involved in this project.

#### 2.1.3.2. Economic Feasibility

Here an evaluation of development cost weighted against the ultimate income or benefit derived from the developed system. The cost for the development of the project has been evaluated and we want to check that the cost does not exceed beneficial cost of the system. The economic and financial analysis is used for evaluating the effectiveness of the candidate system. This project also undergone economic feasibility study and found that it is feasible. So, the cost for development

does not exceed its beneficial cost. This brought to as the conclusion that the system is economically feasible in the context.

#### 2.1.3.3. Operational Feasibility

In operational feasibility the entire application is checked whether the system will be used if it is developed and implemented. Also, it is checked whether there will be resistance from user that may undermine the possible application benefits. There is no barrier for implementing the system. The system also helps to access the information immediately as need arises. Thus, the system is found to be operational feasible.

# SYSTEM SPECIFICATION

## Hardware Specification

The selection of hardware is very important in the existence and proper working of any of the software. When selecting hardware, the size and capacity requirements are also important. The hardware must suit all application developments.

- Processor : i3 or above.
- System Bus : 32 Bit or 64 Bit
- RAM : 4GB or Above
- HDD : 500 GB or Above
- Monitor : 14”LCD or Above
- Keyboard : 108 Keys
- Mouse : Any Type of mouse
- Mobile : Android supported mobile phone

## Software specification

One of the most difficult tasks is selecting software, once the system requirement is find out then we have to determine whether a particular software package fits for those system requirements. This section summarizes the application requirement.

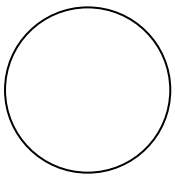
- Operating System : Windows 7or above.
- Technologies used : Android

- Front end : Html,css,bootstrap,JS
- Back End : Python-Django or Flutter/Java
- Database : Mysql
- IDE : PyCharm community , Android studio , PyCharm professional, vs code
- Web browser : Chrome,Explorer,Edge…etc

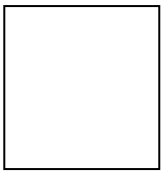
## 3.2 DATA FLOW DIAGRAM

A data flow diagram is graphical tool used to describe and analyze movement of data through a system. These are central tool and the basis from which the other components are developed. The transformation of data from input to output, trough processed, may be described logically and independently of physical components associated with the system. These are known as the logical data flow movement of data between people, departments and workstations. A full description of a system actually consists of a set of data flow diagrams.

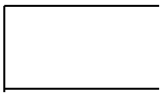
A DFD is also known as a “bubble chart” has the purpose of clarifying system requirements and identifying major transformations that will become programs in system design. So, it is the starting point of the design to the lowest level of detail. A DFD consists of a series of bubbles joined by data flows in the system.
In the DFD, there are four symbols



Process that transforms data flow



Source or Destination of data



Data store

**Rules for drawing data flow diagrams**

Rule 1: Establish the context of the data flow diagram by identifying all of the net input and output data flows

Rule 2: Select a starting point for drawing the DFD.

Rule 3: Give meaningful labels to all data flow lines.

Rule 4: Label all processes with action verbs that relate input and output data flows

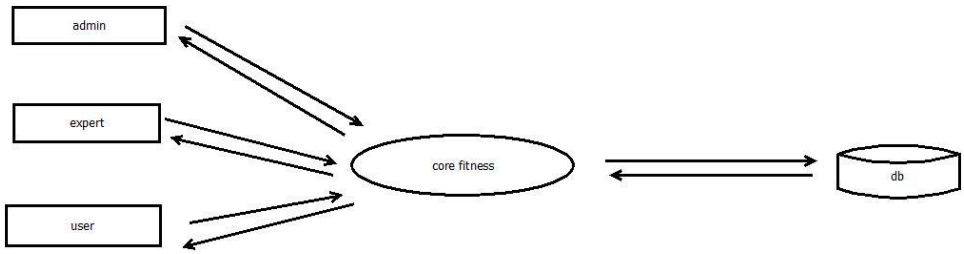
Rule 5: Omit insignificant functions routinely handled in the programming process

Rule 6: Do not include control or flow of control information.

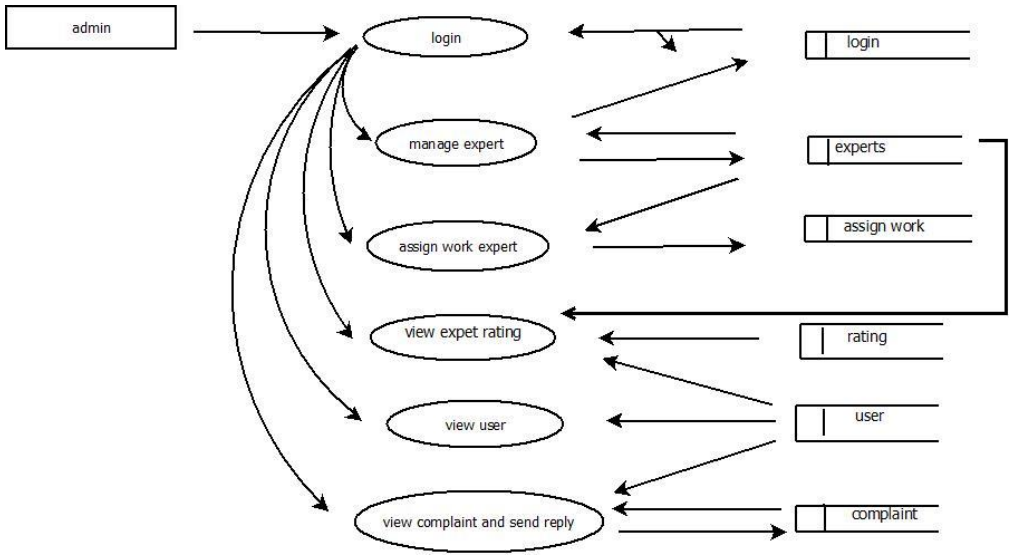
Rule 7: Do not try to put too much information in one DFD.

Rule 8: Be prepared to start over.

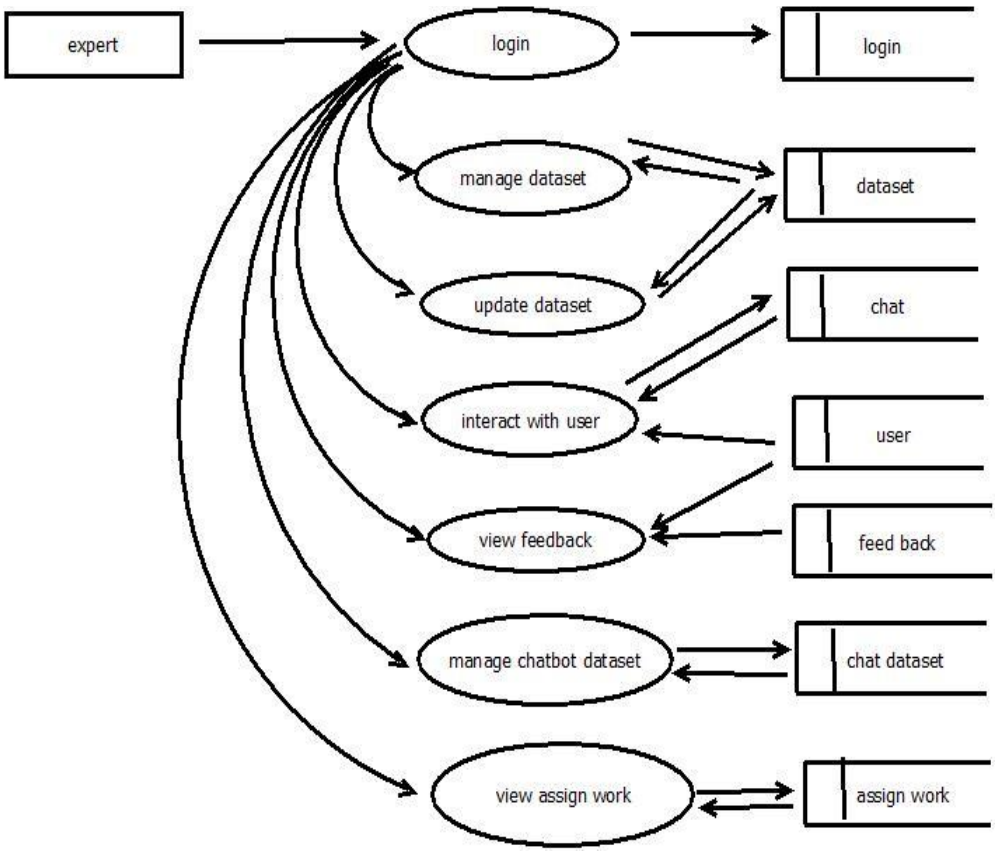
## Level 0



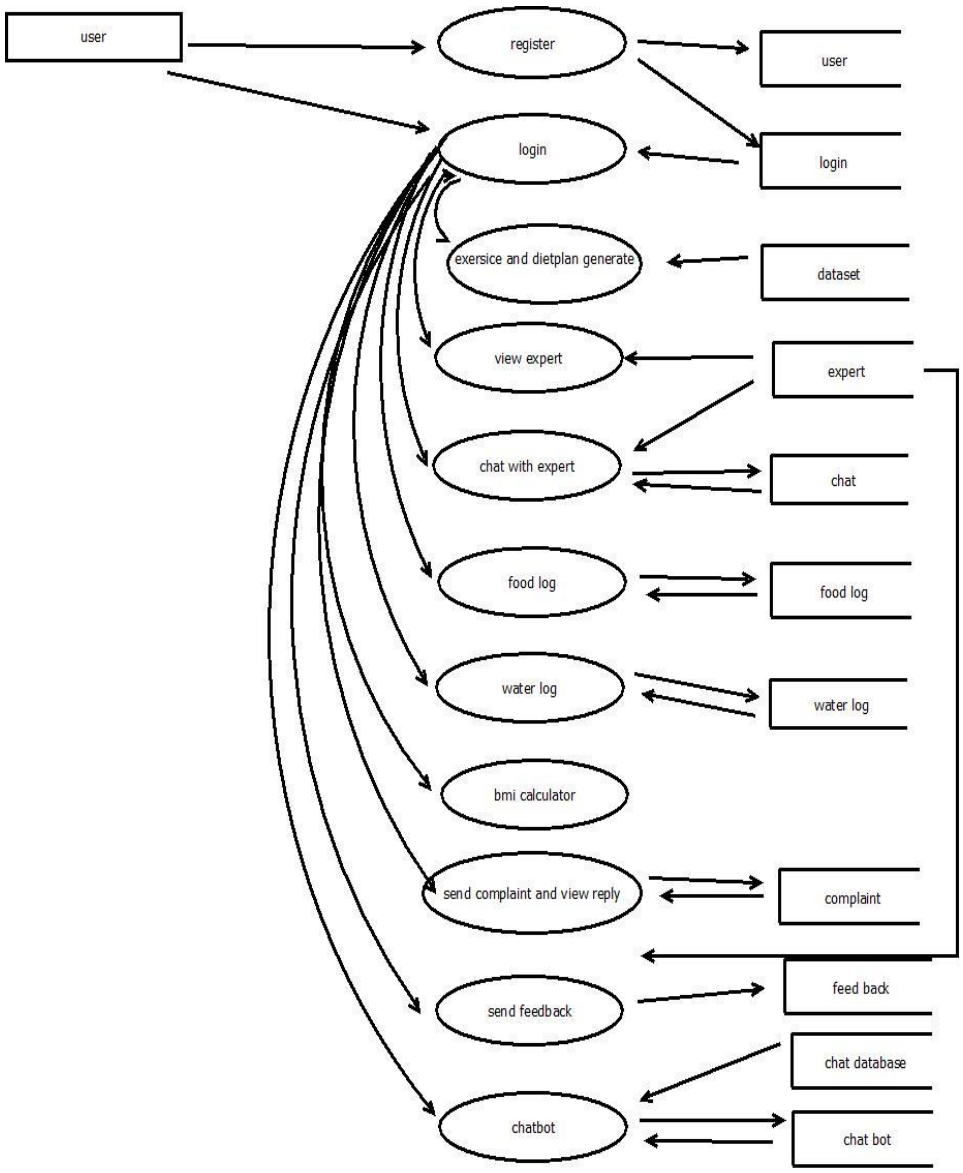

## Level 1 DFD



**Level 2**


expert
login
login
manage dataset
dataset
update dataset
chat
interact with user
user
view feedback
feed back
manage chatbot dataset
chat dataset
view assign work
assign work

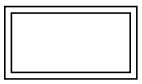
**Level 3**


user
register
user
login
login
exersice and dietplan generate
dataset
view expert
expert
chat with expert
chat
food log
food log
water log
water log
bmi calculator
send complaint and view reply
complaint
feed back
send feedback
chat database
chatbot
chat bot

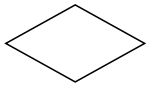
## 3.3 ENTITY RELATIONSHIP DIAGRAM

The ER model is a conceptual data model that views the real world as a construct of entities and associations or relationships between entities. A basic component of the model is the Entity-Relationship diagram, which is used to visually represent data objects. The ER modeling technique is frequently used for the conceptual design of database applications and many database applications and many database design tools employ its concepts.

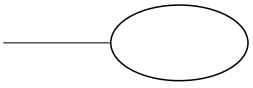
Entity Type



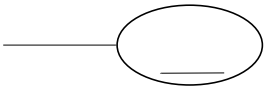
Weak Entity Type



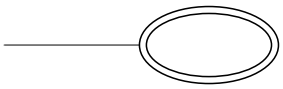
Relationship Type



Attribute

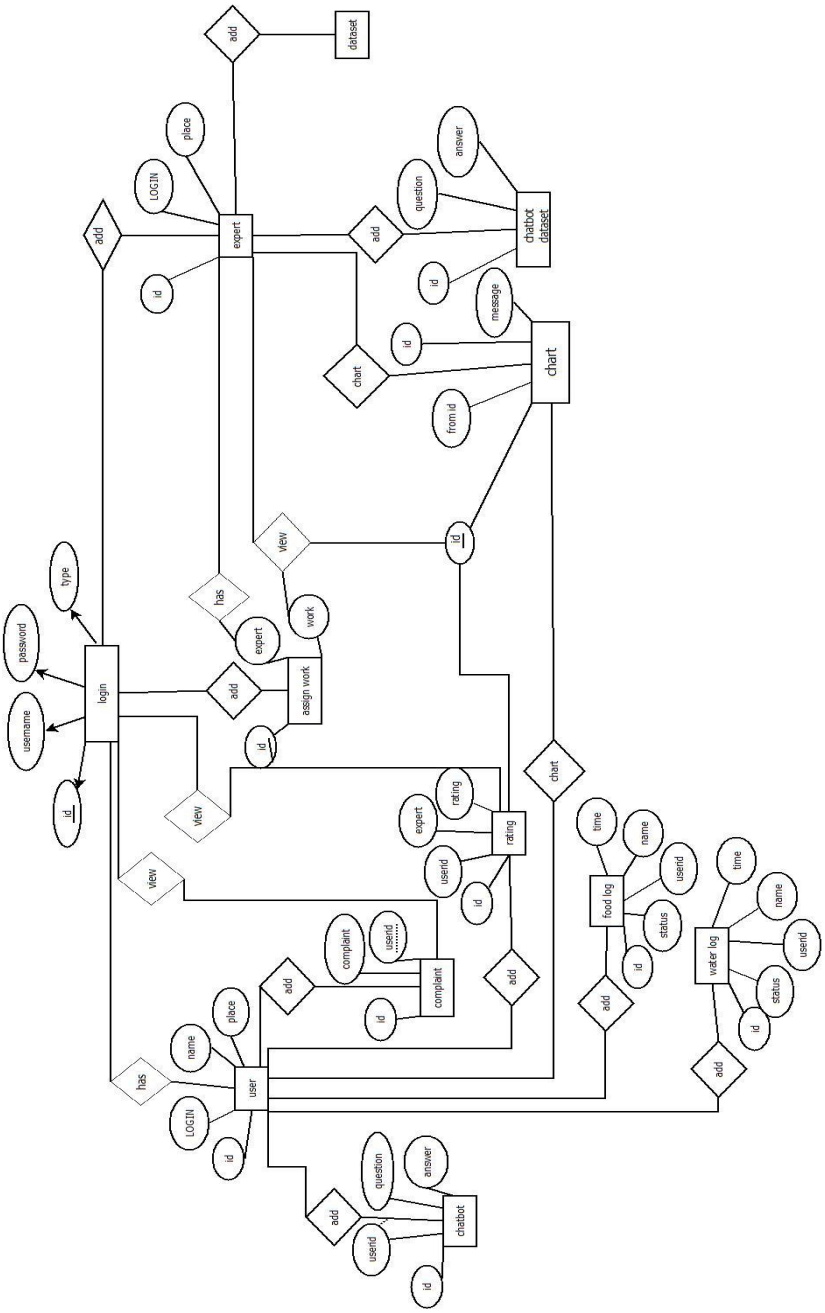


Key attribute



Multivalued Attribute

## ER Diagram Showing Art Gallery Management System



## TABLE DESIGN

Login

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| Username | Varchar(100) | Not null | Username |
| Password | Varchar(100) | Not null | Password |
| type | Varchar(100) | Not null | Type |

EXPERT TABLE

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| Login | Integer | Foreign key | Login |
| First name | Varchar(100) | Not null | First name |
| Last name | Varchar(100) | Not null | Last name |
| Qualification | Varchar(100) | Not null | Qualification |
| Place | Varchar(100) | Not null | Place |
| Post | Varchar(100) | Not null | Post |
| Phone | Big integer | Not null | Phone |
| Email | Varchar(100) | Not null | Email |

**USER TABLE**

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTIION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| Login | Integer | Foreign key | Login |
| First name | Varchar(100) | Not null | First name |
| Last name | Varchar(100) | Not null | Last name |
| Gender | Varchar(100) | Npt null | Gender |
| Place | Varchar(100) | Not null | Place |
| Post | Varchar(100) | Not null | Post |
| Phone | Big integer | Not null | Phone |
| Email | Varchar(100) | Not null | Email |
| Weight | Integer | Not null | Weight |
| Height | Integer | Not null | Height |

ASDESIGN TABLE

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| Expert Id | Integer | Foreign key | Expert Id |
| Work | Varchar(100) | Not null | Work |
| Details | Varchar(100) | Not null | Details |
| Date | Varchar(100) | Not null | Date |
| Status | Varchar(100) | Not null | Status |

CHATBOT DATASET

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| Question | Varchar(100) | Not null | Question |
| Answer | Varchar(100) | Not null | Answer |

COMPLAINT TABLE

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| User Id | Integer | Foreign Key | User Id |
| Complaint | Varchar(100) | Not null | Complaint |
| Date | Varchar(100) | Not null | Date |
| Reply | Varchar(100) | Not null | Reply |

FEEDBACK

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| User Id | Integer | Foreign key | User Id |
| Feedback | Varchar(100) | Not null | Feedback |
| Rating | Float | Not null | Rating |
| Date | Varchar(100) | Not null | Date |
| Expert Id | Integer | Foreign key | Expert Id |

CHAT BOT

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| User Id | Integer | Foreign key | User Id |
| Question | Varchar(100) | Not null | Question |
| Answer | Varchar(100) | Not null | Answer |
| Date | Varchar(100) | Not null | Date |

CHAT TABLE

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| From Id | Integer | Foreign key | From Id |
| To Id | Integer | Foreign key | To Id |
| Message | Varchar(100) | Not null | Message |
| Date | Varchar(100) | Not null | Date |
| Status | Varchar(100) | Not null | Status |

DIET DATASET

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| Bmi | Varchar(100) | Not null | Bmi |
| Age | Varchar(100) | Not null | Age |
| Gender | Varchar(100) | Not null | Gender |
| Bp | Varchar(100) | Not null | Bp |
| Cholestrol | Varchar(100) | Not null | Cholestrol |
| Sugar | Varchar(100) | Not null | Sugar |
| Health Condition | Varchar(100) | Not null | Health Condtion |
| Type | Varchar(100) | Not null | Type |
| Diet Plan | Varchar(100) | Not null | Diet Plan |
| Exercise Plan | Varchar(100) | Not null | Exercise Plan |

WATER LOG

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| User Id | Integer | Foreign key | User Id |
| Alert | Varchar(100) | Not null | Alert |
| Time | Varchar(100) | Not null | Time |
| Status | Varchar(100) | Not null | Status |

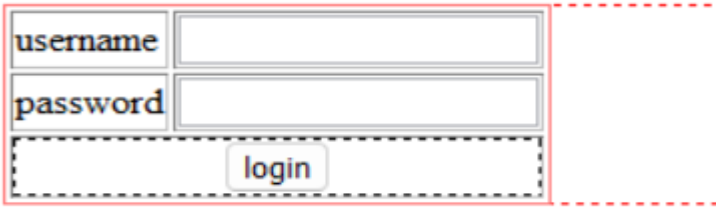
FOOD LOG

| FIELD | DATA TYPE | CONSTRAINS | DESCRIPTION |
|---|---|---|---|
| Id | Integer | Primary key | Id |
| User Id | Integer | Foreign key | User Id |
| Alert | Varchar(100) | Not null | Alert |
| Time | Varchar(100) | Not null | Time |

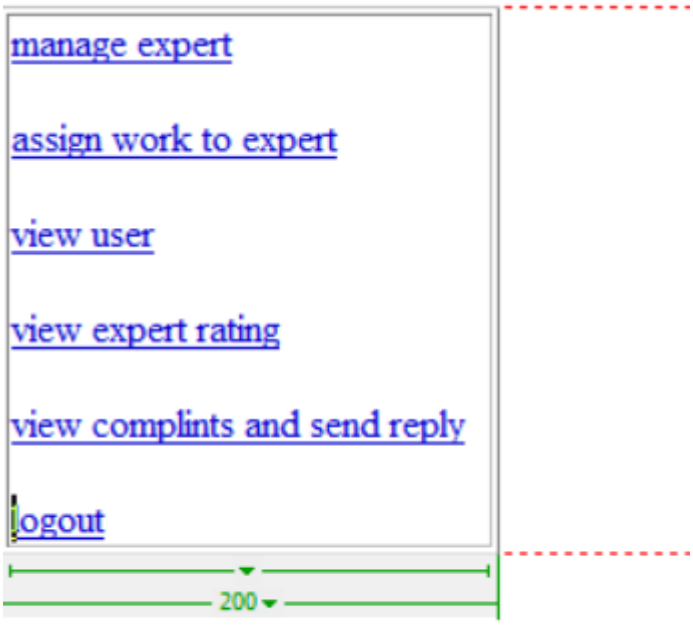
| | | | |
|---|---|---|---|
| Status | Varchar(100) | Not null | Status |

# SYSTEM DESIGN

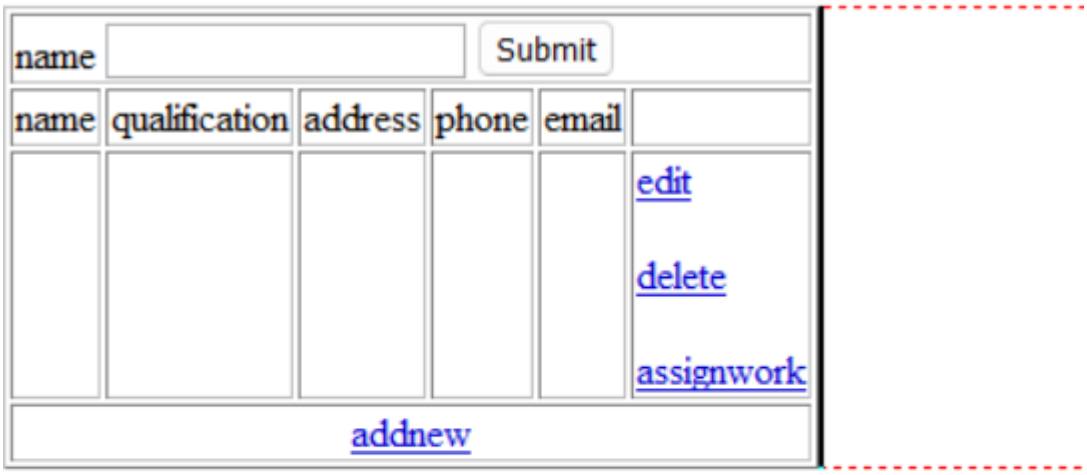
## LOGIN



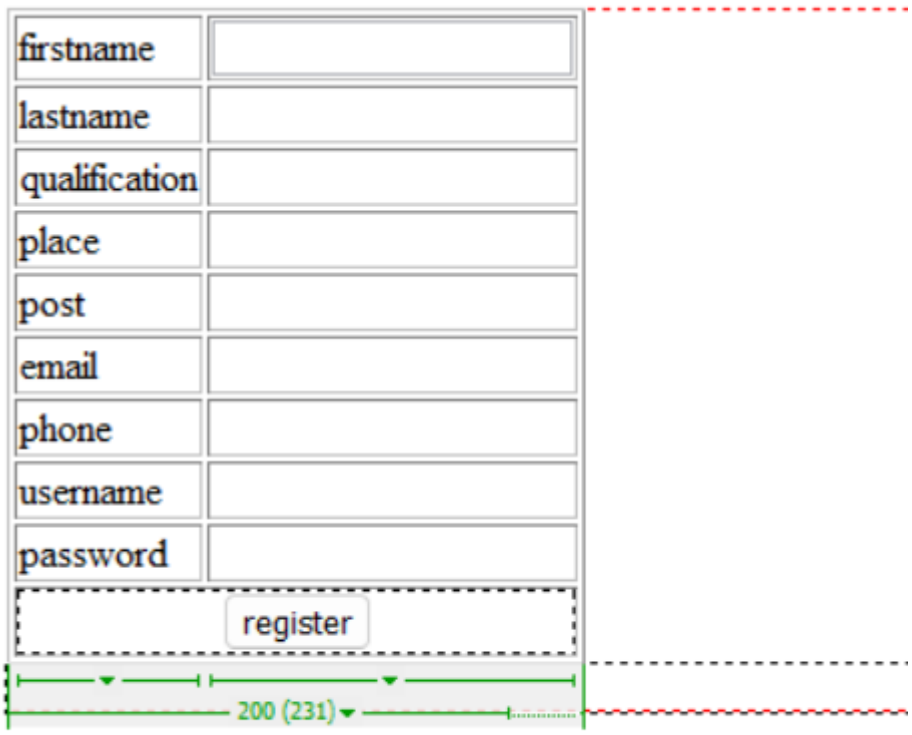

### ADMIN HOME



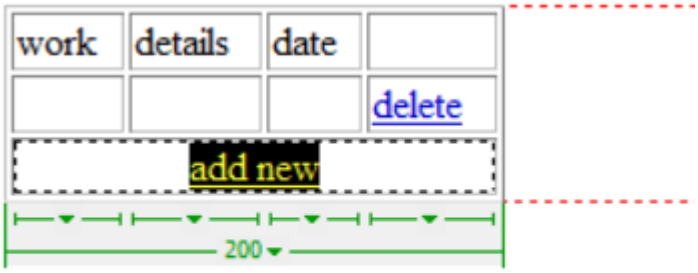

### MANAGE EXPERT



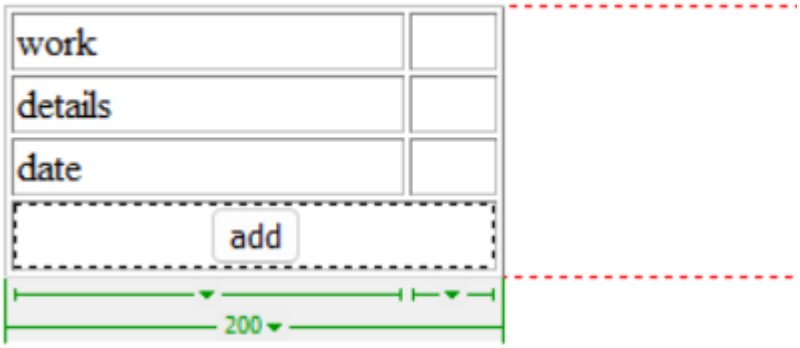

**ADD EXPERT**



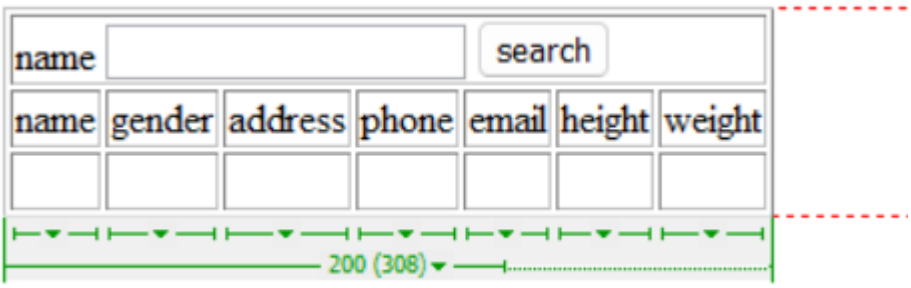

**ASSIGN WORK**



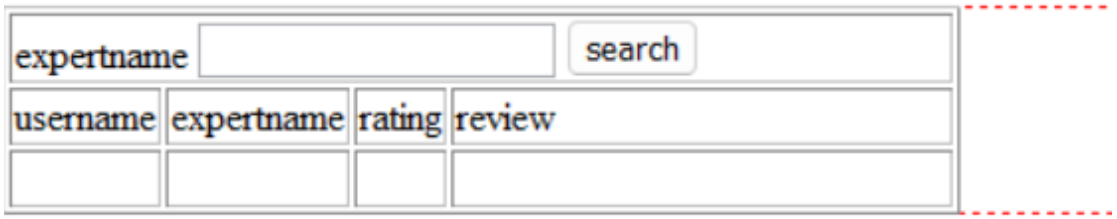

**ADD ASSIGN WORK**




**VIEW USER**

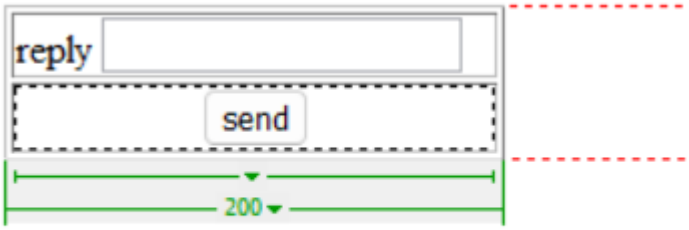



**VIEW RATING**




**VIEW COMPLAINT**

**SEND REPLY**



EXPERT HOME

manage dataset

chat with user

view feedback

manage chatbotdataset

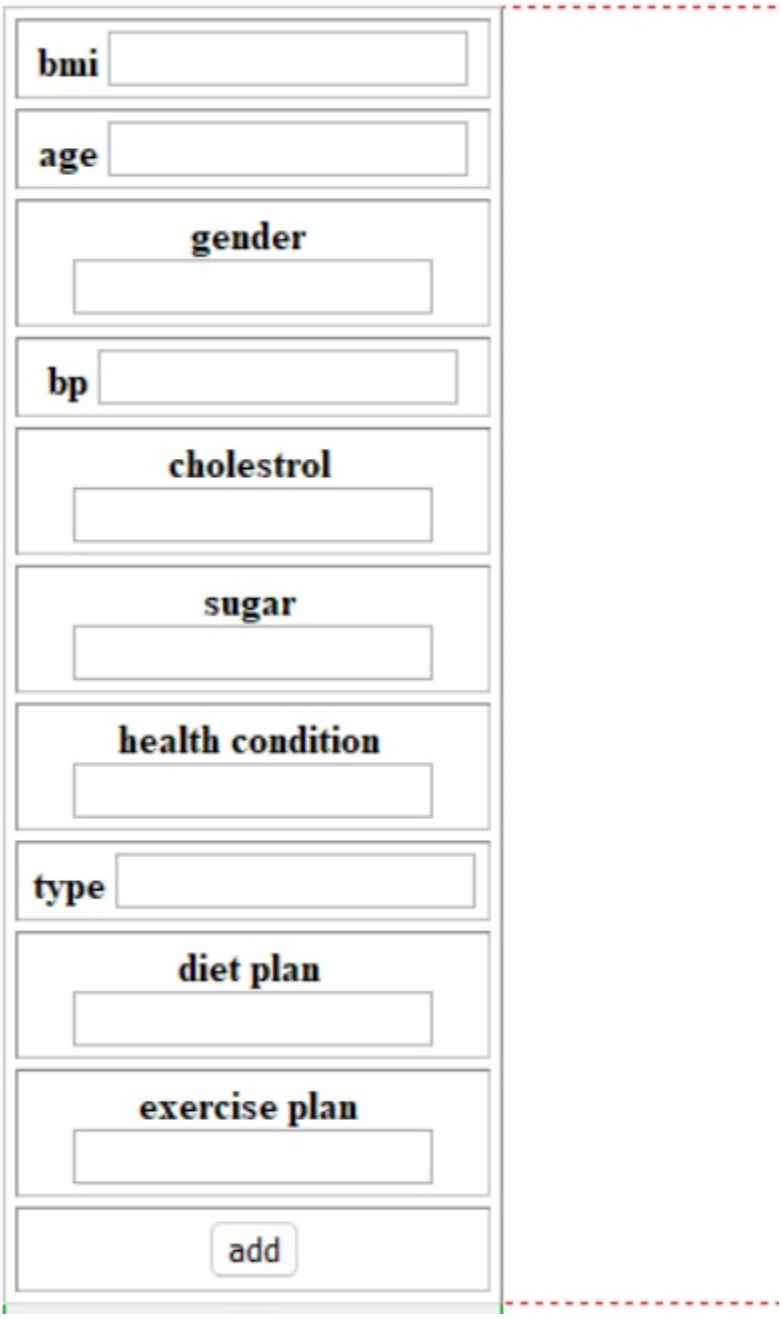
verify assigned work

logout

MANAGE DATASET

| bmi | age | gender | bp | cholestrol | sugar | health condition | type | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | edit<br>delete |
| add new | | | | | | | | | | |

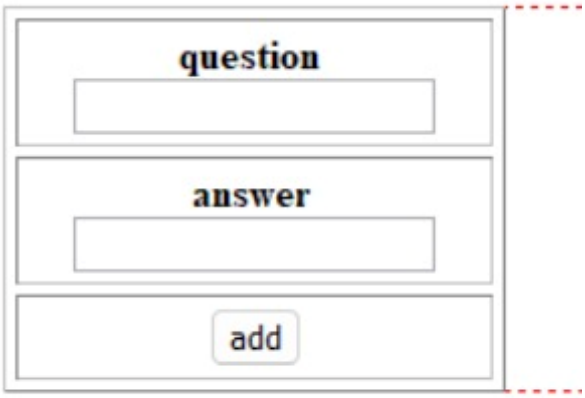
ADD DATASET

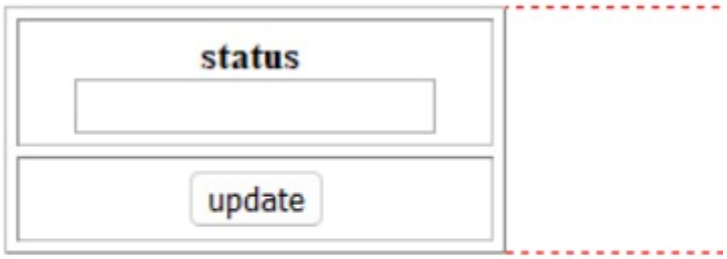

bmi
age
gender
bp
cholestrol
sugar
health condition
type
diet plan
exercise plan
add

VIEW FEEDBACK

MANAGE CHATBOT DATASET

ADD CHATBOT DATASET



VIEW ASSIGNED WORK

UPDATE STATUS